

Embedded Reciprocal Equatives in Mandarin and Semantic Weakening

I-Ta Chris Hsieh

National Tsing Hua University, Taiwan

Abstract

In this paper I present a novel observation on reciprocal equatives (henceforth, RE) in Mandarin: a reciprocal equative, when embedded under a necessity modal or a predicate of desire, gives rise to an unexpected, *at-least-equal-to* interpretation (henceforth, $\geq$ -interpretation). I show that the meaning of a modal sentence or a desire report of this kind may be derived through the postulation of two covert operators: one is akin to *at least* and serves to weaken the meaning of the embedded RE; the other is an exhaustification operator, which gives rise to the inference of sufficiency. Along with Alonso-Ovalle and Hirsch’s (2022) analysis, the proposal draws a connection between the sufficiency modal construction (e.g., *To get good cheese, you only have to go to the North Ends*) and examples of embedded reciprocal equatives investigated in this paper. To the extent that the proposal is on the right track, it provides further support for the postulation of a semantic weakening operator in natural language meaning composition and sheds light on the issue how this weakening operation may be constrained (e.g., Alonso-Ovalle and Hirsch, 2022; Crnič, 2011; Schwarz, 2005).

Keywords— reciprocal equative, weakening, exhaustification, modality, attitude report

1. Introduction

An RE, a degree construction marked with a degree adverb such as Mandarin *yíyàng* (see (1)) or English *equally* (see (2)), expresses the mutual equivalence among the degrees possessed by the objects in comparison along a particular dimension. For instance, both (1) and (2) express that the two people in comparison have the exactly same height¹.

- (1) Zhāngsān hé Lǐsì yíyàng gāo.
Zhangsan AND Lisi the.same tall
‘Zhangsan and Lisi are equally tall.’

- (2) John and Mary are equally tall.

Degree sentences of this kind contrast with English *as*-equatives in that unlike REs, English *as*-

¹Throughout this paper I gloss the coordinator *hé* in the nominal coordination as AND; this is to avoid the complication that *hé*, in some cases, functions like AND and in other cases, like *with* (see Section 2.4). Additionally, the following abbreviations are used when glossing Mandarin examples:

COMP: comparative morpheme; MOD: modification marker; PERF: perfective marker; POSS: possessive marker; RES: resultative marker.

equatives express the $\geq$ -relation between the degrees in comparison rather than mutual equivalence: the *as*-equative in (3) expresses that John's height is **equal to or greater than** Mary's. This is evidenced by the contrast between (3) and (4): unlike an English *as*-equative, continuation of (1)/(2) with *in fact, ZS/John is even taller* simply leads to contradiction.

(3) John is as tall as Mary; in fact, he is even taller.

(4) #Zhāngsān hé Lǐsì yíyàng gāo; shìshíshàng, Zhāngsān shènzhì bǐ Lǐsì gāo.
Zhangsan AND Lisi the.same tall; in.fact, Zhangsan even COMP Lisi tall

Nonetheless, a Mandarin RE unexpectedly gives rise to an $\geq$ -interpretation in two embedding environments, which are discussed in detail below.

1.1. REs under necessity modals

The modalized reciprocal equative (henceforth, modalized RE) like (5) gives rise to an interesting and unexpected puzzle. Consider the scenario and the utterance in (6); the modalized RE uttered by the coach in (6), intuitively, may be paraphrased as (7).

(5) xīnlái-dé duìyuán bìxū hé biě yíyàng gāo.
new.coming-MOD team.player have.to AND Bill equally tall

(6) Context: The basketball team is seeking a new player to replace current shooting guard Bill. While reviewing the profiles of the applicants, the coach says,

xīnlái-dé duìyuán bìxū hé biě yíyàng gāo; zuìhǎo shì nénggòu
new.coming-MOD team.player have.to AND Bill equally tall best FOC possible
bǐ tā hái gāo, qiúduì cái yǒu xīwàng.
COMP 3_{sg} even tall ball.team so.that have hope
'At least the new player and Bill must be equally tall; it would be the best if he could be even taller than Bill; only this way there is hope for the team.'

(7) It is essential for the team that the height of the new player is either the same as Bill's or greater than that.

This intuition, however, does not seem to go along with the standard views in current syntactic and semantic theories. Let's take the usual assumption that modals are propositional operators and at the level of semantic composition (i.e., LF) the subject is interpreted inside their scope (see (8)). In other words, (5) may be assigned an LF representation like (8). Without going into any formal details, the meaning derived from (8) is expected to be like (9).

(8) [ModalP *have.to* [_{vP} [*the-new-player AND Bill*] [_{AP} *equally tall*]]]

(9) Given what the team needs, it is necessary that the new player's height is equivalent to Bill's height.

(9) cannot be equivalent to (7). Suppose that Bill's height is exactly 180cm; an applicant whose height is 185cm would not be considered if the modalized RE in (6) had to be interpreted as (9). On the other hand, this applicant might have a chance if the modalized RE (5) may be interpreted as the paraphrase in (7). Intuitively, the utterance in (6) is not contradictory, and it follows from this utterance that an applicant whose height is greater than Bill's, given the coach's utterance in (6), may be possibly considered. This suggests that the paraphrase in (9) cannot be accurate for the modalized RE in (5), and an analysis along with what has been sketched in (8)-(9) leads to the wrong prediction.

1.2. REs and predicates of desire

A very similar puzzle also arises when an RE is embedded under a predicate of desire, an example of which is given in (10).

- (10) Yūehàn₁-dè māmā xīwàng tā₁ kǎo-dè hé Bǐěr yíyàng gāo.
 John-POSS mother wish he score AND Bill same high

With the usual assumption that a desire predicate operates on the propositional content of its clausal complement, (10) may be parsed at LF, as shown in (11). Accordingly, the interpretation derived may be paraphrased as (12).

- (11) [John's mom [*wish* [_{CP} John and Bill scored equally high]]]

- (12) What John's mom wished for is for John and Bill to score equally high.

Nevertheless, (10) might not always be interpreted this way. Consider (13), a scenario which (10) may be uttered to report the wish of John's mom.

- (13) Bill and John are in the same class. Bill's mom and John's mom are discussing how their kids are doing in school. After checking their reports of the recent exam, John's mom sees that John did not do that well and needs to take a make-up exam. Being worried and wishing that her son could have done better, John's mom expresses to Bill's mom that it would be great if John's grades could be as high as Bill's so that John would not need to take the make-up.

The paraphrase in (12), however, seems too strong to capture this intuition. Along with (12), the wish of John's mom would be satisfied only if John's score and Bill's score were equivalent; therefore, it would be against her wish if John's score were higher than Bill's. This however apparently is not what she wishes for in (13). On the other hand, (14) would be a more appropriate paraphrase to capture the intuition with the desire report (10) when it is uttered to report the wish of John's mom in (13): what she wishes for, in (13), is for John's score to be equal to or even higher than Bill's.

- (14) What John’s mom wishes for is for John’s grades to be equal to Bill’s or even higher.

1.3. Pragmatic effect?

One way to approach the rise of the unexpected $\geq$ -interpretation in the examples above is to see it as a pragmatic effect of some kind: (5) and (10) are parsed as in (8) and (11) and interpreted indeed as (9) and (12). Nevertheless, some mechanism pragmatic might have taken place so that we get the intuition paraphrased as in (7) and (14) respectively.

It seems difficult to pursue an account along these lines. The reason is as follows: generally, the literal meaning of a sentence gets strengthened once some pragmatic mechanism applies. For instance, an implicature that not all students arrived may be pragmatically derived from (15a) (through Grice’s Maxims of Quantity) and therefore the interpretation of (15a) may be strengthened as (15b).

- (15) a. Some students arrived.
b. Some but not all students arrived.

Nevertheless, in the examples we have looked at above, the intuition we get is weaker, rather than stronger, than the interpretation expected: (9) is stronger than the intuition paraphrased in (7), and (12) is definitely no weaker than then intuition paraphrased in (14). This has made it less likely that what has been observed in the data above results pragmatically².

1.4. Outlook and Roadmap

As shown above, a Mandarin RE gives rise to an unexpected $\geq$ -interpretation when it is embedded under a necessity modal or a predicate of desire. Throughout the discussion, I refer to those embedded REs that give rise to such an interpretation as **weak REs**. The goal of this paper hence is to provide a uniform solution to weak REs in these two embedding environments and discuss the implications from these data.

A solution I would like to offer is summarized as follows: in (5) and (10), the complements of the embedding operators (e.g., the modal or the predicate of desire) may contain a covert operator AT-LEAST, whose contribution is akin to *at least* and serves to weaken the meaning of the embedded RE. Additionally, another covert operator EXH akin to *only* is appended on top of these sentences. Along with these lines, (5) and (10) are parsed at LF as in (16a) and (16b) respectively: AT-LEAST gives rise to the $\geq$ -interpretation in these sentences, whereas EXH gives

²Even within the grammatical approach to implicatures (e.g., Fox, 2007, Chierchia et al., 2012; and others), it is difficult to maintain that idea that the $\geq$ -interpretation is the result from additional implicatures generated in the context. As already shown above, an implicature strengthens rather than weakens the meaning of a sentence. See also Section 2.2 for a very brief sketch of such an approach to scalar implicatures.

rise to the intuition that in these examples Bill’s height and Bill’s score is the minimum to reach in order to achieve a particular goal (e.g., in (6)) or satisfy a particular desire (e.g., in (10)).

- (16) a. [EXH-*C'* [ModalP *have.to* [_{vP} AT-LEAST-*C* [*the-new-player and Bill*_F] [_{AP} *equally tall*]]]]
 b. [EXH-*C'* [*John’s mom* [*wish* [_{CP} ... [_{vP} AT-LEAST-*C* [*John and Bill*_F *scored equally high*]]]]]]

The covert operator EXH has been utilized to account for various semantic and pragmatic phenomenon; for instance, scalar implicatures and NPI licensing (see Chierchia et al. (2012), Chierchia (2013) and others). On the other hand, the application of the covert operator AT-LEAST has been suggested in Schwarz, 2005, Crnič, 2011 and Alonso-Ovalle and Hirsch, 2022. Specifically, Alonso-Ovalle and Hirsch (2022) have made use of AT-LEAST to analyze the so called sufficiency modal construction (see (17a) and its LF (17b))³.

- (17) a. To get good cheese, you only have to go to the North Ends.
 b. [*only-C'* [ModalP *have.to* [_{vP} AT-LEAST-*C* [*you go to NE*_F]]]]

Hence, the data under investigation and the analysis sketched above provide additional support for these proposals.

The rest of the discussion is structured as follows. The proposal is laid out in Section 2: in Section 2.1 and 2.2 I illustrate the semantic contribution of the two covert operators AT-LEAST and EXH with the least formalization possible and leave the detailed semantic composition till Section 2.3. Section 3 discusses some implications from the proposal. In Section 4 I present a possible way of regulating the distribution of weak REs.

2. Labor division between weakneing and exhaustification

As already hinted above, it is suggested in this paper that modal statements and desire reports with weak REs (e.g., (5) and (10)) may be parsed with i) AT-LEAST, a covert operator akin to *at least*, in the complement of the modal or the desire predicate, and ii) EXH, another covert operator akin to *only*, on top of the modal sentence or the desire report (see (16a) and (16b)). Both operators are associated with the second coordinate of the nominal coordination (i.e., *y* in (18)).

- (18) [EXH-*C'* ... [□/ATT_{desire} [AT-LEAST-*C* [*x* & [*y*]_F *equally P*]]]]

AT-LEAST weakens the meaning of the RE in its scope so that the embedded RE may give rise

³Alonso-Ovalle and Hirsch’s 2022 analysis will be reviewed in Section 3.1. For other analyses of this construction, see Fintel and Iatridou (2007), Condoravdi and Francez (2022), Alonso-Ovalle and Hirsch (2022) and the references cited therein.

to the $\geq$ -interpretation. On the other hand, EXH ensures that the truth conditions of the whole sentence are not too weak and contributes to the implication that the degree possessed by the second coordinate (e.g., y in (18)) is the minimum that the degree possessed by the first coordinate x should reach in order to achieve a particular goal (as in (5)) or to satisfy the attitude holder's desire (as in (10)). The contribution of these two operators are explained and illustrated in the following two subsections.

2.1. Weakening and the $\geq$ -interpretation

As already shown in (3) and (4), an RE expresses mutual equivalence among the objects in comparison. For simplicity, I paraphrase the meaning of these REs as (19b) and (20b)⁴.

- (19) a. xīnlái-dè duìyuán hé biě yíyàng gāo.
new.coming-MOD team.player AND Bill equally tall
b. HEIGHT(the-new-player)=HEIGHT(Bill)
- (20) a. Yūehàn₁ kǎo-dè hé Biě yíyàng gāo.
John score AND Bill same high
b. SCORE(John)=SCORE(Bill)

The insertion of the covert operator AT-LEAST in the scope of the modal and the predicate of desire, as hinted above, is to weaken the meaning of the embedded RE; as the result, the $\geq$ -interpretation rises under the embedding modal and attitude predicate. In other words, in (16), the modal *have to* and the predicate of desire *wish/hope*, instead of (19b) and (20b), operate on the propositions that may be paraphrased as (21a) and (21b) respectively.

- (21) a. HEIGHT(the-new-player) $\geq$ HEIGHT(Bill)
b. SCORE(John) $\geq$ SCORE(Bill)

The job of AT-LEAST is to take us from (19b)/(20b) to (21a)/(21b). This operator first introduces a set of alternatives to its prejacent, namely, the proposition expressed by its sister at LF. Assume that in (16), AT-LEAST is associated with the second coordinate *Bill* in the embedded RE, all the alternatives in the alternative set C AT-LEAST operates on in (16a) then have the form HEIGHT(the-new-player)=HEIGHT(x) (where given the context of utterance in (6) x can, but need not, be a current team player) and in (16b) SCORE(John)=SCORE(x) (where given the context of utterance (13), x is a student in John's class).

- (22) a. in (16a),
C $\subseteq$ {HEIGHT(the-new-player)=HEIGHT(x): x is an individual}
b. in (16b),

⁴For analyses of reciprocal equatives, see Schwarz, 2007, Nissenbaum and Schwarz, 2011, and Hsieh, 2021.

$$C \subseteq \{\text{SCORE}(\text{John}) = \text{SCORE}(x) : x \text{ is an individual}\}$$

The alternatives in C are rank based on some contextually determined ordering; here we will work with the assumption that for all the alternatives q in C such that q is not identical to the prejacent of AT-LEAST, q is rank higher than the prejacent based on this contextually determined ordering.⁵ For (16a), one plausible way to rank the alternatives is based on the height of individuals the new player is compared to (see (23a), where $\prec$ stands for ‘be rank higher than’); as for (16b), the alternatives may be rank based on the grades of those John is compared to (see (23b)).

(23) For any x and y ,

- a. $[\text{HEIGHT}(\text{the-new-player}) = \text{HEIGHT}(x)] \prec [\text{HEIGHT}(\text{the-new-player}) = \text{HEIGHT}(y)]$ iff $\text{HEIGHT}(x) > \text{HEIGHT}(y)$
- b. $[\text{SCORE}(\text{John}) = \text{SCORE}(x)] \prec [\text{SCORE}(\text{John}) = \text{SCORE}(y)]$ iff $\text{SCORE}(x) > \text{SCORE}(y)$

With the set of alternatives C and the ranking among these alternatives, AT-LEAST then asserts that there is some alternative that is rank at least as high as its prejacent.

(24) AT-LEAST- C p is true iff there is a alternative q in C such that $q \preceq p$ (i.e., q is ranked as high as p) and q is true.

Along with these lines, the $\geq$ -interpretation of the embedded REs in (5) and (10) are derived as follows.

As for (5): with the LF in (16a), the alternative set C in (22a) and the ordering in (23a), the embedded RE with AT-LEAST (i.e., vP in (16a)) asserts that there is an alternative that is rank at least as high as the prejacent and this alternative is true (see (25a)). This paraphrase, along with the idea sketched above, is equivalent to (25b).

- (25) a. There is an alternative q in C such that $q \preceq [\text{HEIGHT}(\text{the-new-player}) = \text{HEIGHT}(\text{Bill})]$ and q is true.
- b. There is some x such that $\text{HEIGHT}(x) \geq \text{HEIGHT}(\text{Bill})$ and $\text{HEIGHT}(\text{the-new-player}) = \text{HEIGHT}(x)$
- c. $\cong \text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(\text{Bill})$

(25b) amounts to saying that the height of the new player is equal to or greater than that of Bill. The $\geq$ -interpretation of the embedded RE in (16a) hence is derived.

As for (10): with the LF in (16b), the alternative set in (22b) and the ordering in (23b), the embedded RE with AT-LEAST asserts that there is some alternative q that is ranked as high as the

⁵See also the formalization in (36a), from Alonso-Ovalle and Hirsch, 2022.

prejacent, namely $\text{SCORE}(\text{John})=\text{SCORE}(\text{Bill})$, and q is true (see (26a)). This, as shown in (26b), amounts to saying that John's score is at least as high as Bill's.

- (26) a. There is an alternative q in C such that $q \preceq [\text{SCORE}(\text{John})=\text{SCORE}(x)]$ and q is true.
b. There is some x such that $\text{SCORE}(x) \geq \text{SCORE}(\text{Bill})$ and $\text{SCORE}(\text{John})=\text{SCORE}(x)$.
c. $\cong \text{SCORE}(\text{John}) \geq \text{SCORE}(\text{Bill})$

2.2. Exhaustification and the inferences of sufficiency

(5) and (10), additionally, carry an inference that the degree possessed by the second coordinate *Bill* is the minimum in order to achieve the goal (in (5), of the team or the coach) or satisfy the desire of the attitude holder (e.g., John's mother in (10)): (5) implicates that the new player need not be taller than Bill in order for him to be considered (see the bold-faced part in (27a)); (10) implicates that John's score need not be higher than Bill's in order for the desire of his mom to be satisfied (see the bold-faced in (27b)).

- (27) a. It is necessary that $\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(\text{Bill})$, and
it is not necessary that $\text{Height}(\text{the-new-player}) > \text{Height}(\text{Bill})$
b. It is in John's mom's wish that $\text{SCORE}(\text{John}) \geq \text{SCORE}(\text{Bill})$, and
it is not in her wish that $\text{Score}(\text{John}) > \text{Score}(\text{Bill})$

The postulation of the covert operator EXH aims to capture this intuition. This operator asserts that its prejacent is true and, at the same time, excludes those alternatives in the alternative set C' not entailed by its prejacent (see (28)).

- (28) EXH- C' p is true iff i) p is true and ii) there is no q in C' such that q is stronger than p and q is true.

Take (29) for instance; *some* may give rise to a *not-all* implicature. Along with Chierchia et al., 2012 and others, this scalar implicature is derived in the way as in (30): *John ate all of the cookies* is an alternative to the prejacent of EXH, namely *John ate some of the cookies*. Suppose that in this case, 'be stronger than' is understood in terms of entailment; namely, q is stronger than p iff q asymmetrically entails p . Given that the *John ate all of the cookies* asymmetrically entails and hence is stronger than the prejacent, it is excluded. The *not-all* implicature hence is derived.

- (29) John ate some of the cookies.

Scalar implicature: John didn't eat all of the cookies.

- (30) a. $[\text{EXH-}C [\text{John ate } [\text{some of the cookies}]_F]]$
b. $C := \{\text{John ate some of the cookies}, \text{John ate all of the cookies}\}$

- c. $\llbracket (30a) \rrbracket = 1$ iff John ate all of the cookies and $\neg$ [he ate all of the cookies]

Back to the case of weak REs; this is how EXH does its job: first, in (16), EXH introduces a set alternatives C' to its prejacent (i.e., via C' in (16a) and (16b)). In the discussion below, I use $\text{NECESSARY}(p)$ as the shorthand for *it is necessary that p* and $\text{WISH}_{j-m}(p)$ for *John's mom wishes p*.

- (31) a. In (16a),
 $C' \subseteq \{\text{NECESSARY}(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(x)) : x \text{ is an individual}\}$
 b. In (16b),
 $C' \subseteq \{\text{WISH}_{j-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(x)) : x \text{ is an individual}\}$

These alternatives, just like those in (22), are ranked with respect to a contextually given ordering, and one plausible candidate for the basis of this ordering is the height for (16a) and the grades for (16b), just as in (23)⁶. Along with these lines, the alternatives EXH operates on in (16) are ranked in the way as in (32).

- (32) For any x and y ,
 a. $[\text{NECESSARY}(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(x))] \prec [\text{NECESSARY}(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(y))]$ iff $\text{HEIGHT}(x) > \text{HEIGHT}(y)$
 b. $[\text{WISH}_{j-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(x))] \prec [\text{WISH}_{j-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(y))]$ iff $\text{SCORE}(x) > \text{SCORE}(y)$

As for (5): with the LF in (16a), the set of alternatives in C' in (31a) and the ordering in (32a), the application of the operator EXH gives rise to the truth conditions in (33).

- (33) a. $\text{NECESSARY}(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(\text{Bill}))$, and
 b. there is no q in C' such that q is true and q is stronger than $\text{NECESSARY}(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(\text{Bill}))$.
 c. $\cong$ For any x such that $\text{HEIGHT}(x) > \text{HEIGHT}(\text{Bill})$, $\neg \text{NECESSARY}(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(x))$

(32a) says that it is necessary for the new player's height to be equivalent to or greater than Bill's. On the other hand, EXH excludes the alternatives stronger than its prejacent and gives rise to (32b), which amounts to saying that there is no need for John's height to be greater than

⁶For convenience and simplicity in presentation, I assume that EXH and AT-LEAST are interpreted against the same ordering on alternatives. This assumption on EXH is consistent with Coppock and Beaver, 2013, according to which in an *only*-sentence, the ordering of alternatives is based on a pragmatically determined scale. Nothing would change if it were assumed that the ordering of the alternatives is determined by asymmetric entailment (e.g., as in (30)).

Bill's (i.e., it is permissible that $\text{HEIGHT}(\text{the-new-player}) = \text{HEIGHT}(\text{Bill})$). This is the sufficiency inference observed.

As for (10): with the LF (16b), the set of alternatives C' in (31b) and the ordering in (32b), the application of EXH leads to the truth conditions in (34).

- (34) a. $\text{WISH}_{j-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(\text{Bill}))$, and
 b. there is no q in C' such that q is true and q is stronger than $\text{WISH}_{j-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(\text{Bill}))$.
 c. $\cong \text{For any } x \text{ such that } \text{SCORE}(x) > \text{HEIGHT}(\text{John}),$
 $\neg \text{WISH}_{j-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(x))$

In these truth conditions, (34a) says that John's mom wishes that John's score be equal to or greater than Bill's; (34b), which is derived through the application of EXH, says that her wish need not be satisfied by John's score being higher than Bill's (i.e., she would be okay with John's score being equivalent to Bill's). This captures the intuition that Bill's score would be the minimum for John's to reach if the wish of John's mom were to be satisfied.

2.3. Formalization and composition

Along with the illustrations presented in the two subsections above, this subsection is devoted to the formalization of these ideas. For convenience, I take a reciprocal equative like (1) to express the equivalence among the objects in comparison along the dimension in question. Therefore, the embedded reciprocal equatives in (5)/(16a) and (10)/(16b) translate as the propositions in (35a) and (35b) respectively.

- (35) a. $\lambda w. \llbracket \text{the-new-player and Bill } yíyàng \text{ tall} \rrbracket^w =$
 $\lambda w. \text{HEIGHT}_w(\text{the-new-player}) = \text{HEIGHT}_w(\text{Bill})$
 b. $\lambda w. \llbracket \text{John and Bill score } yíyàng \text{ high} \rrbracket^w =$
 $\lambda w. \text{SCORE}_w(\text{John}) = \text{SCORE}_w(\text{Bill})$

The denotation of the covert operator AT-LEAST, following Alonso-Ovalle and Hirsch, 2022, is given in (36): it carries a presupposition that all the other alternatives are rank higher and the prejacent, and as shown in (24), AT-LEAST then asserts that there is an alternative q such that q is rank as high as the prejacent and q is true.

- (36) a. $\llbracket \text{AT-LEAST} \rrbracket^{w, \preceq^S} = \lambda C_{\langle \langle s, t \rangle, t \rangle}. \lambda p_{\langle s, t \rangle}. \forall q \in C[p \neq q \rightarrow q \prec_{sp}]. \exists q \in C[p \preceq_{sp} \text{ and } q(w)]$
 b. $q \prec_{sp}$ iff q is ranked higher than p with respect to a contextually determined ordering S .

The denotation of the covert operator EXH is given in (37); as already shown in (28), this operator contributes semantically in a way similar to *only*: it excludes all the alternatives that are ranked higher than the prejacent, given the ordering S' .

$$(37) \quad \llbracket \text{EXH} \rrbracket^{w, \preceq_{S'}} = \lambda C_{\langle \langle s, t \rangle, t \rangle} \cdot \lambda p_{s, t} \cdot p(w) \text{ and } \neg \exists q \in C [q(w) \text{ and } q \prec_{S'} p]$$

As for (5): with the LF in (16a) and the alternative set C in (38), vP in (16a) is assigned the truth conditions in (39). Assuming that the contextually determined ordering S is based on the height of the alternatives to Bill, as shown in (40), these truth conditions amount to saying that the new player's height is as great as Bill's (see (25)).

$$(16) \quad \text{a.} \quad [\text{EXH-} C' [_{\text{ModalP}} \text{have.to} [_{vP} \text{AT-LEAST-} C [\text{the-new-player and Bill}_F] [_{AP} \text{equally tall}]]]]$$

$$(38) \quad C \subseteq \{ \lambda w_s. \text{HEIGHT}_w(\text{the-new-player}) = \text{HEIGHT}_w(x) : x \subseteq D_e \}$$

$$(39) \quad \begin{aligned} \llbracket vP \rrbracket^{w, \preceq_S} \text{ is defined iff:} \\ \text{for all } q \in C \text{ s.t. } q \neq [\lambda w. \text{HEIGHT}_w(\text{the-new-player}) = \text{HEIGHT}_w(\text{Bill})], \\ q \prec_S [\lambda w. \text{HEIGHT}_w(\text{the-new-player}) = \text{HEIGHT}_w(\text{Bill})]; \\ \text{if defined, } \llbracket vP \rrbracket^{w, \preceq_S} = 1 \text{ iff:} \\ \text{there is some } q \in C \text{ s.t. } q(w) = 1 \text{ and} \\ q \preceq_S [\lambda w. \text{HEIGHT}_w(\text{the-new-player}) = \text{HEIGHT}_w(\text{Bill})] \end{aligned}$$

$$(40) \quad \text{Given } x \in D_e, y \in D_e, \text{ and } w \in D_s:$$

$$\begin{aligned} \text{if } \text{HEIGHT}_w(x) > \text{HEIGHT}_w(y), \text{ then} \\ [\lambda w_s. \text{HEIGHT}_w(\text{the-new-player}) = \text{HEIGHT}_w(x)] \prec_S \\ [\lambda w_s. \text{HEIGHT}_w(\text{the-new-player}) = \text{HEIGHT}_w(y)] \end{aligned}$$

Assuming that a necessity modal such as *have to* quantifies over all the accessible worlds, the covert operator EXH then operates on the proposition expressed by ModalP in (16a), namely that it is necessary that $\text{HEIGHT}(\text{the-new-player}) = \text{HEIGHT}(\text{Bill})$ (see (41)). Given the set of alternatives C' and the contextually determined ordering S' (see (42)-(43)), (16a), after the application of EXH, is assigned the truth conditions in (44)⁷.

$$(41) \quad \text{a.} \quad \llbracket \text{have.to} \rrbracket^w = \lambda p_{\langle s, t \rangle} \cdot \forall w' \in \text{ACC}_w. p(w')$$

⁷ As a reviewer points out, it entails that in all worlds quantified over x is more than y that in all worlds quantified over $x=y$. Nevertheless, this has nothing to do with the core problem with respect to the interpretation of embedded REs discussed here. As already discussed in length above, the problem we are facing with respect to the embedded REs is that while the modal and attitude reports with embedded REs are expected to be true IFF in all worlds where $x=y$, they may be true in situations where only the weak truth conditions, namely that in all worlds quantified over, are verified.

- b. $\llbracket \text{ModalP} \rrbracket^w = 1$ iff:
for all $w' \in \text{ACC}_w$: there is some $q \in C$ s.t. $q(w') = 1$ and
 $q \leq_S [\lambda w'. \text{HEIGHT}_{w'}(\text{the-new-player}) = \text{HEIGHT}_{w'}(\text{Bill})]$
- c. $\cong$ For all $w' \in \text{ACC}_w$: $\text{HEIGHT}_{w'}(\text{the-new-player}) \geq \text{HEIGHT}_{w'}(\text{Bill})$
(shorthand: $\Box_w(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(\text{Bill}))$)

$$(42) \quad C' \subseteq \{ \lambda w_s. \Box_w(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(x)) : x \in D_e \}$$

$$(43) \quad \text{Given } x \in D_e, y \in D_e, \text{ and } w \in D_s:$$

if $\text{HEIGHT}_w(x) > \text{HEIGHT}_w(y)$, then
 $[\lambda w'. \Box_w(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(x))] <_S$
 $[\lambda w'. \Box_w(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(y))]$

$$(44) \quad \llbracket (16a) \rrbracket^{w, \preceq_S} = 1 \text{ iff:}$$

- a. $\Box_w(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(\text{Bill}))$, and
b. there is no $q \in C'$ s.t.
 $q <_S [\lambda w. \Box_w(\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(\text{Bill}))]$ and $q(w)$
c. $\cong \neg \Box_w(\text{HEIGHT}(\text{the-new-player}) > \text{HEIGHT}(\text{Bill}))$

As already noted earlier, the exclusive component of these truth conditions (see (44c)), amount to saying that the new player's height need not be greater than Bill's. Together with (44a), this gives rise to the inference that Bill's height is the minimum for that of the new player to reach (see also (33)).

As for (10): with the LF in (16b) and the alternative set in (45), vP in (16b) is assigned the truth conditions in (46) (see also (23b)). As shown in (47), these propositional alternatives are ranked based on the score (in the world of evaluation w) of the individuals alternative to Bill. These truth conditions then amount to saying that John's score is as high as Bill's height.

$$(16) \quad \text{b. } [\text{EXH-}C' [{}_S \text{ John's mom } [\text{wish } [{}_{CP} \dots [{}_{vP} \text{ AT-LEAST-}C [\text{John and Bill}_F \text{ scored equally high }]]]]]]]$$

$$(45) \quad C \subseteq \{ [\lambda x_e. \lambda w. \text{SCORE}_w(\text{John}) = \text{SCORE}_w(x)] : x \in D_e \}$$

$$(46) \quad \llbracket vP \rrbracket^{w, \preceq_S} \text{ is defined iff:}$$

for all $q \in C$ s.t. $q \neq [\lambda w. \text{SCORE}_w(\text{John}) = \text{SCORE}_w(\text{Bill})]$,
 $q <_S [\lambda w. \text{SCORE}_w(\text{the-new-player}) = \text{SCORE}_w(\text{Bill})]$;

if defined, $\llbracket vP \rrbracket^{w, \preceq_S} = 1$ iff:

there is some $q \in C$ s.t. $q(w) = 1$ and $q \leq_S [\lambda w. \text{SCORE}_w(\text{John}) = \text{SCORE}_w(\text{Bill})]$

(47) Given $x \in D_e$, $y \in D_e$, and $w \in D_s$:

if $\text{SCORE}_w(x) > \text{SCORE}_w(y)$, then

$$[\lambda w. \text{SCORE}_w(\text{John}) \geq \text{SCORE}_w(x)] <_S [\lambda w. \text{SCORE}_w(\text{John}) \geq \text{SCORE}_w(y)]$$

One crucial ingredient needed in the next step is a semantics for the attitude predicate *xīwàng* ‘wish’. I assume the one in (48), following Fintel (1999): *wish* is interpreted with a modal base $f_i(w)$ and an ideal $g_i(w)$; it asserts that its complement is true in all worlds w' such that w' is in $f_i(w)$ and w' conforms with the attitude holder’s desire in the word of evaluation w , which is described by $g_i(w)$ (see (48))^{8,9}.

(48) $\llbracket \text{wish}_i \rrbracket^{w,f,g} = \lambda p_{\langle s, t \rangle} \cdot \lambda x_e$:

- (i) $\text{DOX}_{x,w} \cap p = \emptyset$; (i.e., x ’s belief is incompatible with p)
 - (ii) $\text{DOX}_{x,w} \subseteq f_{i,x}(w)$; (i.e., x ’s belief is included in the modal base)
 - (iii) $f_{i,x}(w) \cap p \neq \emptyset$; (i.e., the modal base is compatible with p)
 - (iv) $f_{i,x}(w) - p \neq \emptyset$. (i.e., the modal base is compatible with $\neg p$)
- $\forall w' \in \text{BEST}_{g_{i,x}(w)}(f_{i,x}(w))$: $w' \in p$;

where $f_{i,x}(w)$ and $g_{i,x}(w)$ are the modal base and the ordering source that are anchored at the attitude holder x and world of evaluation w .

The modal base and the ordering source of *wish* are subject to the constraints that are taken to be presuppositions in (48-i)-(48-iv); for detailed discussions on these presuppositions, I refer the reader to Fintel (1999); given that these components are not pertinent to the discussion below, I will put them aside for simplicity. Along with (48), the constituent S in (16b) is assigned the truth conditions in (50), which amount to saying that in the wish of John’s mom, John’s score is at least as high as Bill’s.

(49) a. $\llbracket S \rrbracket^{w,f,g} = 1$ iff:

for all $w' \in \text{BEST}_{g_{i,J's-m}(w)}(f_{i,J's-m}(w))$: there is some $q \in C$ s.t. $q(w') = 1$ and

⁸In this Kratzerian-style semantics for *wish* by Fintel (1999), the kinds of modal base function f and the ordering source function g chosen for *wish* is specified lexically through the index i . Technically, f_i is a function from worlds to sets of worlds; g_i is a function from worlds to sets of propositions. In (48), $g_{i,x}(w)$ is the set of propositions that describe x ’s desires in w . The so called ‘best’ worlds (i.e., the worlds that best conform with x ’s desires in w) are chosen through the strict partial ordering in (i, b).

- i. a. $g_{i,x}(w) = \{p_{\langle s, t \rangle} : p \text{ describes a desire of } x \text{ in } w\}$
- b. (i) $w' \leq_{g_{i,x}(w)} w''$ (namely, w' is at least as good as w'' with respect to $g_{i,x}(w)$) iff for all $p \in g_{i,x}(w)$, if $w'' \in p$, then $w' \in p$
- (ii) $w' <_{g_{i,x}(w)} w''$ (namely, w' is better than w'' with respect to $g_{i,x}(w)$) iff $w' \leq_{g_{i,x}(w)} w''$ and $\neg(w'' \leq_{g_{i,x}(w)} w')$
- (iii) $\text{BEST}_{g_{i,x}(w)}(f_{i,x}(w)) = \{w' : w' \in f_{i,x}(w) \text{ and } \neg \exists w'' [w'' \in f_{i,x}(w) \text{ and } w'' <_{g_{i,x}(w)} w']\}$

⁹The semantics of *wish* in (48), without the definedness condition (48-i), may be extended to the predicate of desire *hope*.

- $q \preceq_S [\lambda w' : \text{SCORE}_{w'}(\text{John}) = \text{SCORE}_{w'}(\text{Bill})]$
- b. $\cong$ for all $w' \in \text{BEST}_{g_i, J's-m}(w) : \text{SCORE}_{w'}(\text{John}) \geq \text{SCORE}_{w'}(\text{Bill})$
 (shorthand: $\text{WISH}_w^{J's-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(\text{Bill}))$)

The next step is to incorporate the exclusive component by applying EXH. As noted above, EXH operates on the set of alternatives C' (see (50)) to its prejacent, namely the proposition *John's mom wishes that John's score is equal to or greater to Bill's score*. Crucially, all the alternatives in C' are ranked based on the ordering in (51).

$$(50) \quad C' \subseteq \{ \lambda w_s. \text{WISH}_w^{J's-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(x)) : x \in D_e \}$$

$$(51) \quad \text{Given } x \in D_e, y \in D_e, \text{ and } w \in D_s:$$

if $\text{SCORE}_w(x) > \text{SCORE}_w(y)$, then

$$\begin{aligned} & [\lambda w'. \text{WISH}_w^{J's-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(x))] \prec_S \\ & [\lambda w'. \text{WISH}_w^{J's-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(y))] \end{aligned}$$

EXH asserts that its prejacent, namely *John's mom wishes that John's score were equal to or greater to Bill's*, is true and excludes all the alternatives in C' that are ranked higher than its prejacent. This gives to the truth conditions in (52). The exclusive component amounts to saying that it is not in the wish of John's mom that John's score were greater than Bill's. Together with the prejacent, this further amounts to saying that Bill's score is the minimum to satisfy John's mom's wish.

$$(52) \quad \llbracket (16b) \rrbracket^{w, \preceq_S} = 1 \text{ iff:}$$

- a. $\text{WISH}_w^{J's-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(\text{Bill}))$, and
- b. there is no $q \in C'$ s.t. $q \prec_S [\lambda w. \text{WISH}_w^{J's-m}(\text{SCORE}(\text{John}) \geq \text{SCORE}(\text{Bill}))]$ and $q(w)$
- c. $\cong \neg \text{WISH}_w^{J's-m}(\text{SCORE}(\text{John}) > \text{SCORE}(\text{Bill}))$

2.4. AT-LEAST and *at least*

In the analysis presented above, the postulation of AT-LEAST plays a crucial role. One therefore might wonder how this assumption may be empirically motivated within the domain of investigation. Given that AT-LEAST is assumed to be the covert counterpart of *at least*, it is then expected to find parallelism between these two. This, as shown below, is evidenced by the contrasts between (6) and (53a) and between (10) and (54).

$$(53) \quad \text{The same scenario as in (6) ...}$$

- a. #xīnlái-dé duìyuán hé bǐěr bīxū yíyàng gāo.
 new.coming-MOD team.player AND Bill have.to equally tall

- (54) Yūehàn₁-dè māmā xīwàng tā₁ hé Bǐěr kǎo-dè yíyàng gāo.
John-POSS mother wish he AND Bill score same high

(53a) and (6) differ only in word order, and so do (54) and (10). In (6) and (10), the first coordinate, namely *the new player* in (6) and *Bill* in (10), are located in subject position leaving the rest of the nominal coordination behind. On the other hand, in (53a) and (54), the whole nominal coordination is in subject position staying intact. This word order difference has effects in interpretation: uttering (53a) in the context in (6) is intuitively infelicitous, which suggests that (53a) does not have the $\geq$ -interpretation. Likewise, (54) only has the meaning that John’s mom wishes for John’s score and Bill’s score to be equivalent, rather than for John’s score to be at least as high as Bill’s.

In Mandarin, the first coordinate of a nominal coordination may be separated from the rest of the coordination without leading to any change in meaning; both (55a) and (55b) say that Zhangsan and Lisi went to Taipei yesterday¹⁰. This is extended to REs; while (56a) contrasts with (56b) in word order, both express that Zhangsan and Lisi had the same height last year.

- (55) a. Zhāngsān hé Lǐsì zuótiān qù-lè táiběi.
Zhangsan AND Lisi yesterday go.to-PERF Taipei
b. Zhāngsān zuótiān hé Lǐsì qù-lè táiběi.
Zhangsan yesterday AND Lisi go.to-PERF Taipei.
‘Zhangsan and Lisi went to Taipei yesterday.’
- (56) a. Zhāngsān hé Lǐsì qùnián yíyàng gāo.
Zhangsan AND Lisi last.year the.same tall
b. Zhāngsān qùnián hé Lǐsì yíyàng gāo.
Zhangsan last.year AND Lisi equally tall
‘Zhangsan and Lisi were equally tall last year.’

To the best of my knowledge, there are two different views on the split of nominal coordination in Mandarin. In one (e.g., Paris, 2008 *et seq*), Mandarin *hé* may be lexically ambiguous between English *and* and *with*. The first coordinate may be separated from rest of the coordination phrase when it is used as English *with*. In this case, the coordinator *hé* together with the second coordinate forms a constituent PP which excludes the first coordinate (see (57b)).

- (57) Paris (2008)
- a. (55a): [Z AND L] *yesterday* [_{vP} [~~Z-AND-L~~] go to Taipei] ($\cong$ English *and*)
b. (55b): *Z yesterday* [_{vP} ~~Z~~ [AND L] [*go-to Taipei*]] ($\cong$ English *with*)

On the other hand, Zhang (2010) argues that the split of the nominal coordination in. e.g., (55a),

¹⁰This is not totally accurate. It has been reported in Zhang, 2010 that when the first coordinate is separated from the rest of the coordination (e.g., (55b) and (56b)), only the collective interpretation is available. Given that this does not affect the discussion below, I put this aside for convenience and simply refer the reader to Zhang, 2010 for details.

simply results from the movement of the first coordinate out of the coordination phrase, as shown in (58).

- (58) Zhang (2010)
- a. (55a): [*Z* AND *L*] *yesterday* [_{vP} [~~*Z* AND *L*~~] go to Taipei]
- b. (55b): *Z yesterday* [_{vP} [~~*Z*~~ AND *L*] go to Taipei]

In the analysis of weak REs presented above, the second coordinate of the nominal coordination is in focus associated with the operator AT-LEAST, which is located at Spec *v*P. Along with this assumption together with either of the analyses of coordination above, the unavailability of the $\geq$ -interpretation of embedded REs in (53) and (54) may be attributed to the failure of AT-LEAST to be associated with the second coordinate: given that in these two examples, the second coordinate is structurally higher than AT-LEAST, it cannot be associated with AT-LEAST. The $\geq$ -interpretation hence cannot arise. The reasoning above also provides an explanation why the embedded English REs' in (59)-(60) do not give rise to an $\geq$ -interpretation: in both examples, the second coordinate in the nominal coordination is structurally higher than AT-LEAST; therefore focus association with AT-LEAST fails and consequently the $\geq$ -interpretation cannot arise.

- (59) a. John and Bill need to be equally tall (in order for John to be considered).
 b. $*[\text{EXH } [\text{need } [J\mathcal{E}[B]_{\text{F}} [\text{AT-LEAST } \cancel{J\mathcal{E}B} \text{ be equally tall }]]]]]$
- (60) a. I wish that John's score and Bill's were equally high.
 b. $*[\text{EXH } I \text{ wish that } [J's \text{ score } \mathcal{E} B_{\text{F}}'s \text{ score } [\text{AT-LEAST } \cancel{J's \text{ score } \mathcal{E} B's \text{ score}} \text{ be equally high}]]]$

It is worth noting that the same restriction is also observed with overt VP-*at least*. As shown in (61a) and (62), while the second coordinate may be associated with overt *at least* when the first coordinate is separated from the rest of the coordination (61a)/(62a), (61b)/(62b) shows that when the nominal coordination is located in subject position intact, the second coordinate cannot be associated with *at least*.

- (61) a. Wángwǔ zuótiān zhìshǎo hé sān_F-gè xuéshēng tāolùn zhè-jiàn shì.
Wangwu yesterday at.least AND 3-CL student discuss this-CL matter
'Yesterday Wangwu discussed this matter with at least three students.'
- b. *Wángwǔ hé sān_F-gè xuéshēng zuótiān zhìshǎo tāolùn zhè-jiàn shì.
Wangwu AND 3-CL student yesterday at.least discuss this-CL matter
Intended: 'Wangwu and at least three students discussed this matter yesterday.'
- (62) a. Zhāngsān shàng-gè-yuè zhìshǎo hé Lǐsǐ_F jiàn-guò-miàn.
Zhangsan previous-CL-month at.least AND Lis meet-EXP
'Last month Zhangsan met with at least Lisi_F.'

- b. *Zhāngsān hé Lǐsì shàng-gè-yuè zhìshǎo jiàn-guò-miàn.
 Zhangsan AND Lisi previous-CL-month at.least meet-EXP
 Intended: ‘Zhangsan and at least Lisi met last month.’

This contrast suggests that the association with VP-*at least* needs to be rightward rather than leftward at the surface, just like what we have seen with AT-LEAST. While the exact account for this restriction has to be left for further investigation, the parallelism we have seen above does empirically motivate the postulation of AT-LEAST in weak REs: given that AT-LEAST is simply the covert counterpart of *at least*, it is expected to see where AT-LEAST is postulated the effect from the constraints *at least* is subject to. And this is indeed what we have seen in the word order restriction on weak REs.

Excursion

As a reviewer points out, an explanation along with the lines above might beg for the question why reconstruction of the subject cannot ameliorate the failure of focus association with *at least*/AT-LEAST in the examples discussed above. As the discussion below shows, it depends on the lexical nature of the focus particle whether reconstruction of the subject may rescue rightward focus association. As pointed out by Jackendoff (1972), association with *even* can be rightward, but this is not the case for *only*. Further investigation is required to see why *at least*/*At-least* patterns with *only* but not *even*; this would take us too far from the center of the discussion in this paper and hence would have to be left for future investigation.

The word order restriction on association with AT-LEAST/*at least* shown above is reminiscent of Jackendoff’s (1972) observation on focus association with VP-*only*; as shown in (63), from Jackendoff (1972, pp. 248-250), VP-*only* must c-command its associate; hence, the association with VP-*only* can only be rightward.

- (63) a. *[John]_F *only* gave his daughter a new bicycle.
 b. John *only* [gave]_F his daughter a new bicycle.
 c. John *only* gave [his]_F daughter a new bicycle.
 d. John *only* gave his [daughter]_F a new bicycle.
 e. John *only* gave his daughter a [new]_F bicycle.
 f. John *only* gave his daughter a new [bicycle]_F.

This suggests that the effect we have seen with AT-LEAST/*at least* may follow from a general constraint on focus association, though the exact nature of this constraint would have to be left for future research.

Jackendoff (1972) additionally points out that focus association with *even*, unlike *only*, is not subject to this constraint.

- (64) a. John_F *even* gave his daughter a new bicycle. (Jackendoff, 1972, p. 248)

- b. John_F will *even* gave his daughter a new bicycle. (Erlewine, 2014, p. 132, (10))

Erlewine (2014) provides an account for this restriction on VP-*only* along with the Copy Theory *à la* Chomsky (1993) and Fox (1999, 2002), a.o, though it is unclear how this account may be extended to *at least*. In a nutshell, he suggests that associating VP-*only* with the low copy of its associate leads to contradictory truth conditions. I simply refer the reader interested to Erlewine (2014) for details. .

3. Implications and remarks

3.1. Connection to sufficiency modal constructions

As already hinted above, the account laid out above is built on Alonso-Ovalle and Hirsch’s (2022) analysis of sufficiency modal constructions (see (17a)). Intuitively, (17a) says that the North Ends is one of the place where good cheese are available, and compared to other options, going there is the easiest way to achieve this goal.

- (17) a. To get good cheese, you only have to go to the North Ends.

Sufficiency modal constructions have been discussed in Alonso-Ovalle and Hirsch, 2022; Condoravdi and Francez, 2022; Fintel and Iatridou, 2007; Krasikova, 2010; and others. Among these work, Fintel and Iatridou (2007) first point out that this modal construction poses what they call the Prejacent Problem for the common assumption on *only*, according to which in an *only*-sentence *only* ϕ (e.g., *Only John ate broccoli*), the truth of the prejacent ϕ (e.g., in the example aforementioned, *John ate broccoli*) is presupposed rather than asserted (Horn, 1969; and others). Along with this assumption, (17a) (with the parse (65a)) presupposes that to get good cheese, you have to go to the North Ends.

- (65) a. [*only* [*you have to go to the* [North Ends]_F]]
 b. Presupposition: You have to go the North Ends (to get good cheese).

This, however, is inconsistent with the intuition; (17a) does not at all say that there is any particular place where one needs to go to get good cheese. In addition, as pointed out by Alonso-Ovalle and Hirsch (2022), this problem cannot be solved by weakening the presuppositional component in the meaning of *only* to an existential one (i.e., *only* ϕ presupposes that there is an alternative to ϕ that is true; see Horn, 1996 and others): with the existential presupposition from *only*, (17a) presupposes that there is some place one needs to go to get good cheese, which is still inconsistent with the intuition.

In their account, Alonso-Ovalle and Hirsch (2022) suggest that the Prejacent Problem may

be solved by positing in the scope of the modal a weakening operator such as AT-LEAST (see (66) and the denotation of AT-LEAST in (36)). Suppose that the options to get good cheeses are the North End (NE), New York City (NYC) and Switzerland (CH). With the set of alternatives C and the ordering (which correlates with the effort to get to these places) in (67), the truth conditions of vP_2 in (66) (see (68)) amount to saying that you go to NE or NYC or CH. Consequently, the truth conditions of vP_1 amount to saying that it is required that you go to NE or NYC or CH.

$$(66) \quad [_{TP} \text{ only-}C' [_{vP_1} \text{ have.to } [_{vP_2} \text{ AT-LEAST-}C [you \text{ go to the } [North \text{ Ends}]_F]]]]]]$$

$$(67) \quad \begin{array}{l} \text{a. } C=\{[\lambda w_s. \text{ you go to } x \text{ in } w]: x=NE, NYC, \text{ or } CH\} \\ \text{b. } [\lambda w_s. \text{ you go to } x \text{ in } NE] \prec_S [\lambda w_s. \text{ you go to } x \text{ in } NYC] \prec_S [\lambda w_s. \text{ you go to } x \text{ in } CH] \end{array}$$

$$(68) \quad \begin{array}{l} \llbracket vP_2 \rrbracket^{w, \preceq_S} \text{ is defined iff:} \\ \quad \text{for all } q \in C \text{ s.t. } q \neq [\lambda w'_s. \text{ you go to } NE \text{ in } w'], q \prec_S [\lambda w_s. \text{ you go to } NE \text{ in } w]; \\ \quad \text{if defined, } \llbracket vP_2 \rrbracket^{w, \preceq_S} = 1 \text{ iff:} \\ \quad \text{there is some } q \in C \text{ s.t. } q(w)=1 \text{ and } q \preceq_S [\lambda w'_s. \text{ you go to } NE \text{ in } w'] \\ \quad \cong \text{You go to NE or NYC or CH.} \end{array}$$

Given the presence of AT-LEAST, instead of (69), vP_1 in (66) is assigned the weaker truth conditions in (70), which amount to saying that (to get good cheese) you are required to go to one of NE, NYC and CH.

$$(69) \quad \text{for all } w' \in Acc_w: \text{ you go in } w' \text{ to NE}$$

$$(70) \quad \begin{array}{l} \llbracket vP_1 \rrbracket^{w, \preceq_S} = 1 \text{ iff for all } w' \in Acc_w: \text{ you go in } w' \text{ to NE or NYC or CH} \\ \text{(Simplified as: } \Box_w (NE \vee NYC \vee CH)) \end{array}$$

Even with the lexical meaning of *only* as in (71) (Horn, 1969; a.o.), according to which *only* ϕ presupposes that ϕ is true, vP_2 , with the truth conditions (70) rather than (69), is presupposed to be true. The Prejacent Problem therefore is solved.

$$(71) \quad \llbracket \text{only} \rrbracket = \lambda C_{\langle \langle s, t \rangle, t \rangle}. \lambda p_{\langle s, t \rangle}: p(w). \text{ for all } q \in C \text{ such that } q(w), p \subseteq q$$

For completeness of the discussion, the contribution of *only* in (17) and the truth conditions of this modal sentence are illustrated as follows. With the set C' of alternatives as in (72a), the truth conditions of (17)/(66) are derived as (72b): the presupposition says that you are required to go to NE or NYC or CH. The assertion, on the other hand, says that you are not required to go to NYC or CH, which with the assumption that the ordering of these alternatives correlates with the efforts to get there, amount to saying that you are not required to go further than NE.

- (72) a. $C' = \{\Box_w(\text{NE} \vee \text{NYC} \vee \text{CH}), \Box_w(\text{NYC} \vee \text{CH}), \Box_w(\text{CH})\}$
 b. $\llbracket (17)/(66) \rrbracket^{w, \prec^s}$ is defined iff $\Box_w(\text{NE} \vee \text{NYC} \vee \text{CH})$;
 if defined, for all $q \in C'$ s.t. $q(w)$, $q \subseteq \Box_w(\text{NE} \vee \text{NYC} \vee \text{CH})$
 $\cong \neg \Box_w(\text{NYC} \vee \text{CH})$ and $\neg \Box_w(\text{CH})$

As noted above, the exhaustification operator EXH is seen as akin to covert *only* (Chierchia, 2004, 2013; Chierchia et al., 2012; Fox, 2007; a.o.). The analysis of weak REs proposed above therefore may be seen as a straightforward extension of Alonso-Ovalle and Hirsch’s (2022) analysis of sufficiency modal constructions: (5)/(16a) are parsed in the same way as (17)/(66), with *only* being replaced with EXH.

- (16) a. $[\text{EXH-}C' [\text{ModalP } \textit{have.to} [\text{vP } \textit{AT-LEAST-}C [\textit{the-new-player and Bill}_F] [\text{AP } \textit{equally tall}]]]]$

Given that the postulation of the operator AT-LEAST accounts for weak REs under a desire predicate as well, these data lend further support for the appeal to an operator like AT-LEAST to capture the weakening phenomenon in general in natural language meaning.

As a reviewer points out, a sufficiency modal construction like (17) carries an implication of easiness: it implicates that going to the North Ends, compared to other options, is an ‘easy’ means to achieve the designated goal, namely getting good cheese. On the other hand, modal statements and attitude reports with weak REs do not carry such an implication; for instance, (5) intuitively does not implicate that Bill is ‘minimum tall player’ by any standard.

Indeed, an embedded RE like (5), unlike a sufficiency modal construction, does not implicate that Bill’s height is counted as ‘low’ for ‘tall players’, and this is consistent with the analysis proposed above. This difference between sufficiency modal constructions and embedded REs may have to do with the difference between *only* and the covert operator EXH: while ‘only’ may be used with what Fintel and Iatridou (2007) called ‘diminishing’ function even outside sufficiency modal constructions, this doesn’t seem to be so for EXH: while (73) may implicate that being a soldier is ranked ‘low’ on the relevant scale, (74) does not seem to carry an implication that the number of cookies John ate is low.

- (73) He is only a soldier. (Fintel and Iatridou, 2007, (124))

- (74) John ate some cookies.

Additionally, as argued extensively in Fintel and Iatridou, 2007, implications of easiness from a sufficiency modal construction is not a truth-conditional entailment but something like an implicature from this construction; therefore, such an implication should not be encoded in the lexical meaning of *only*. Consequently, the lack of implications of ‘easiness’ shall not undermine the possi-

ble connection between a sufficiency modal construction like (17) and modal statements with weak REs like (5).

3.2. The alternative of obtaining alternatives?

The proposal laid out above dwells on the assumption that the alternatives AT-LEAST and EXH operates on are constructed based on the second coordinate of the nominal coordination in the weak RE: in (16a) and (16b), the set of alternatives are constructed based on the second coordinate *Bill*. Alternatively, one may obtain the alternative set based on the comparison relation expressed under the modal or the embedding predicate, as illustrated in (75): in (16a), AT-LEAST operates on the alternative set in (75a); in (16b), it operates on the alternative set in (75b).

$$(75) \quad \begin{array}{ll} \text{a.} & \text{in (16a), } C := \left\{ \begin{array}{l} \text{HEIGHT(the-new-player)} > \text{HEIGHT(Bill)}, \\ \text{HEIGHT(the-new-player)} = \text{HEIGHT(Bill)} \end{array} \right\} \\ \text{b.} & \text{in (16b), } C := \left\{ \begin{array}{l} \text{SCORE(John)} > \text{SCORE(Bill)}, \\ \text{SCORE(John)} = \text{SCORE(Bill)} \end{array} \right\} \end{array}$$

With these alternative sets, we derive the same truth conditions already shown above. Take (16a) for instance; along with (75a), the weak RE in (16a) amounts to saying that $\text{HEIGHT(the-new-player)} \geq \text{HEIGHT(Bill)}$. After exhaustification, the same truth conditions are derived that it is necessary that $\text{HEIGHT(the-new-player)} \geq \text{HEIGHT(Bill)}$ and it is not necessary that $\text{HEIGHT(the-new-player)} > \text{HEIGHT(Bill)}$ (see (27a)/(44c)).

Plausible as it might initially seem to be, however, further scrutinization shows that an alternative analysis along these lines may potentially lead to the wrong prediction. Recall from Section 2.4 that the $\geq$ -interpretation disappears when the nominal coordination in the RE as a whole moves to sentential subject position (see (53a) and (54)); consequently, the second coordinate in these examples is structurally higher than AT-LEAST. With the proposal above, this maybe attributed to the independent constraint that focus association with AT-LEAST/*at least* cannot be rightward. On the other hand, with the way of constructing the alternative set as in (75), there seems no way to distinguish (53a) and (5) at LF, for these two examples are parsed the same way at LF; this then renders it difficult to accommodate the word order effect we have seen in Section 2.4.

Conceptually the way of constructing alternatives in (75) is less preferred. As pointed out in Fox and Katzir, 2011; Katzir, 2007, and many others, although relevance plays a role, the construction of alternatives is structurally constrained. In a nutshell, the set of alternatives to a given sentence S is obtained from replacing focused constituents within S with constituents that are at most as complex as the original constituents. The proposal laid out in Section 2 is consistent with such a view: having the second coordinate in focus, the alternatives AT-LEAST and EXH operate on are obtained from replacing the semantic object denoted by the second coordinate with the one

denoted by some constituent that could be at most as complex. Nevertheless, the sets of alternatives like (75) are inconsistent with such a view, for in Mandarin an RE and a comparative of superiority differ not only in the degree morpheme but also in the syntactic structure, and hence (75) cannot be obtained simply by replacing the RE morpheme *yíyàng* with the comparative morpheme *bǐ* even if *yíyàng* is in focus. The structure of the *bǐ*-comparative may still be controversial and subject to further discussion (see Erlewine, 2018; Hsieh, 2015, 2017 and others); it is however clear that unlike an RE like (76a) a *bǐ*-comparative may not involve a nominal coordination structure in subject position. For the *bǐ*-comparative (76b) to be generated as an alternative to the RE (76a), some morpho-syntactic transformations need to apply that are unauthorized given current syntactic and semantic theories; for instance, the removal of the coordination structure in the subject position. All these possible morpho-syntactic complexities then make analyses that appeal to an alternative set like (75) unwarranted¹¹.

- (76) a. Zhāngsān hé Lǐsì yíyàng gāo. (RE)
 Zhangan AND Lisi equally tall.
 ‘Zhangsan and Lisi are equally tall.’
 b. Zhāngsān bǐ Lǐsì gāo. (comparative of superiority)
 Zhangsan COMP Lisi tall
 ‘Zhangsan is greater than Lisi.’

4. Weakening and scalarity

An embedded RE does not always carry the $\geq$ -interpretation. Taking AT-LEAST to be the source of this, this suggests that the occurrence of AT-LEAST needs to be constrained. Outside goal-oriented modal sentences and desire reports we have seen above, an RE may give rise to the $\geq$ -interpretation as well when it is embedded in the antecedent of a conditional, as shown in (77). In its preferred interpretation, (77) does not say that Bill would have been admitted if his score and Bill’s were exactly the same; instead, it expresses that Bill would have been admitted if his score were equal to or greater than John’s.

- (77) rúguǒ Bǐer shùxué kǎo-dè hé Yuēhàn yíyàng gāo, ta zǎo jiù jìn
 if Bill mathematics score.RES AND John equally high he already then enter
 táidà lè.
 Taiwan.University PERF
 ‘If Bill scored as high as John in mathematics, he would have entered Taiwan University.’

Nevertheless, we have also seen that an unembedded RE does not give rise to such an interpretation, as already shown in (4). Additionally, such an interpretation does not arise in all kinds of attitude reports: while a weak RE may be seen under a predicate of desire, this seems to be not the case for predicates of belief (e.g., *believe* and *think*). For instance, (78), where an RE is embedded under

¹¹I thank a reviewer for reminding me of the need for a more detailed discussion on this issue.

believe/think, can only express that in the belief of John’s mom, John’s score and Bills are equal.

- (78) Yūehàn-dè māmá xiāngxìn/rènwéi Yūehàn kǎo-dé hé Bǐěr yíyàng gāo.
 John-POSS mother believe/think John score-RES AND Bill equally high
 ‘John’s mom thinks that John and Bill scored equally high.’

At this moment a proper characterization of such a constraint is not clear yet. Nevertheless, there seemingly exists some clues that suggest that the distribution of weak REs is parallel to that of what Crnič (2011) called weak *even* (i.e., the occurrence of *even* such that the minimal clause it is located at surface structure denotes a proposition that is most likely among the alternatives): (79) shows that weak *even* is infelicitous in unembedded contexts. (80) shows that in contrast to desire reports, weak *even* is infelicitous in belief reports. Although conditionals are not discussed in Crnič, 2011, (81) shows that weak *even* is licensed in the antecedent of a conditional.

- (79) #John made even ONE vide of that quality.
- (80) a. #Mary believes John found even ONE party that cares for the people.
 b. I hope that John makes even ONE video of that quality. (Crnič, 2011, 65, (1a))
 c. This Haley chick wishes she could sing even ONE note close to Janis.
 (Crnič, 2011, 66, (4b))
- (81) If John had made even ONE video of that quality, he would have been invited to join this project.

The parallelism we have seen above strongly suggests that Crnič’s (2011) analysis of weak *even* might be a viable option to regulate the distribution of weak REs.

4.1. Weak *even* and preferences

Crnič’s (2011) account is built on the assumptions in (82)¹²:

- (82) a. *even* triggers a scalar presupposition that its prejacent (i.e., the proposition it operates on) needs to be less likely than some alternative in the alternative set (Bennett, 1982; Kay, 1990).

¹²Karttunen and Peters (1979) suggest a semantics of *even* according to which the scalar presupposition of *even* is universal rather than existential.

- i. a. $\llbracket \text{even} \rrbracket^{w,c}(C_{\langle(s,t),t\rangle})(p_{\langle s,t \rangle})$ is defined only if $\forall q \in C[q \neq p \rightarrow p \triangleleft_c q]$
 b. $p_{\langle s,t \rangle} \triangleleft_c q_{\langle s,t \rangle}$ iff p is less likely than q given the likelihood provided in c .

As noted by Crnič (2011), the choice of theses two views does not affect the discussion here.

- b. *even* may move out of its base position at LF (Karttunen and Peters, 1979; Lahiri, 1998; and others).
 - c. The likelihood *even* is interpreted against needs to be compatible with the constraint in (83c) (Crnič, 2011, 26, (3)).
 - d. The desiderative component in the semantics of predicates of desire and the relevant preference in the context may play a crucial role in rendering the scalar presupposition of *even* plausible. There is, however, no such a component in that of a predicate of belief that could serve for this purpose (Crnič, 2011, p. 96).
- (83) a. $\llbracket \textit{even} \rrbracket^{w,c} = \lambda C_{\langle \langle s, t \rangle, t \rangle} \cdot \lambda p_{\langle s, t \rangle} \cdot \exists q \in C[p <_c q] \cdot p(w)$
- b. $p <_c q$ iff p is less likely than q given a likelihood determined in c .
- c. Scalarity and entailment
- If a proposition p entails a proposition q , q cannot be less likely than p .

Take (79) for instance. *even* in this example triggers a scalar presupposition that the prejacent of *even*, namely that *John made one video of that quality*, is less likely than some alternative *John made n video of that quality* in the alternative set (see (84)). This scalar presupposition however fails and consequently weak *even* is infelicitous here: given that *John made n video of that quality* entails *John made one video of that quality*, the prejacent of *even* cannot be less likely than any alternatives in the alternative set.

$$(84) \quad C \subseteq \{ \textit{John made } n \textit{ videos of that quality} : n \geq 1 \}$$

Along with a Kratzerian-style modal semantics of attitude predicates *à la* Fintel, 1999 (e.g., (48) for *wish* and *hope*), Crnič (2011) suggests that a desire report with weak *even* in the complement may be parsed as (85a): EXH first exhaustifies the alternatives to its prejacent *John hopes to make one video of that quality*. *even* then moves at LF over the predicate of desire *hope* and triggers the scalar presupposition that its prejacent is less likely than some alternative in C' . Along with Crnič, 2011, the prejacent of *even* may be paraphrased as *John hopes to make one video of that quality and he is okay with making exactly one video of that quality*. Below I use the shorthand $\Box_{\text{hope}}(\text{one}) \wedge \Diamond_{\text{hope}}(\text{!one})$ to present the prejacent of *even* in this example.

- (85) a. $\llbracket \textit{even-C'} \rrbracket [\text{EXH-C} [\textit{John}_1 \textit{ hope} [\text{PRO}_1 \textit{ make ONE}_F \textit{ video of that quality}]]]$
- b. $C' \subseteq \{ \Box_{\text{hope}}(n) \wedge \Diamond_{\text{hope}}(\text{!n}) : n \geq 1 \}$
- c. $\llbracket (85a) \rrbracket^w$ is defined only if:
- $\exists q \in C' [\Box_{\text{hope}}(\text{one}) \wedge \Diamond_{\text{hope}}(\text{!one}) <_c q];$ (Scalar presupposition)
- if defined, $\llbracket (85a) \rrbracket^w = 1$ iff John hopes to make 1 video of that quality and he is okay with making exactly 1 video of that quality in w .

As noted by Crnič (2011), the scalar presupposition in (85c) may be satisfied in a context c if in c it is taken for granted that making more than one video of that quality is more desirable to John than making exactly one video. With a likelihood ordering based on the preference of this kind, (85c) may be satisfied in such a context since there is an alternative in C' that is more likely than the prejacent (i.e., John making more than one video is more likely to satisfy his desire than him making exactly one video). Consequently, weak *even* may be felicitous under a predicate of desire in such a context.

In (87a), *even* operates on an exhausted belief report, which is presented with the shorthand $\Box_{\text{DOX}}(\text{one}) \wedge \Diamond_{\text{DOX}}(!\text{one})$ and paraphrased as *John believes that he made one video of that quality and it seems to him that he made exactly one video of that quality*. As shown in (87b), the scalar presupposition of *even* amounts to saying that it is less likely than some alternative in C' that John believes that he made 1 video of that quality and it seems to him that he made exactly 1 video of that quality.

$$(86) \quad \llbracket \text{think/believe}_i \rrbracket^{w,f,g} = \lambda p_{\langle s, t \rangle} \cdot \lambda x_e: f_{i,x}(w) = \text{DOX}_{x,w}. \forall w' \in f_{i,x}(w): p(w') = 1$$

- (87) a. $\llbracket \text{even-}C' \llbracket \text{EXH-}C \llbracket \text{John}_1 \text{ believe } \llbracket \text{he}_1 \text{ make ONE}_F \text{ video of that quality } \rrbracket \rrbracket \rrbracket$
 b. $\llbracket (87a) \rrbracket^w$ is defined only if:
 $\exists q \in C' [\Box_{\text{DOX}}(\text{one}) \wedge \Diamond_{\text{DOX}}(!\text{one}) \triangleleft_c q];$ (Scalar presupposition)
 if defined, $\llbracket (85a) \rrbracket^w = 1$ iff John believes that he made 1 video of that quality and it seems to him that he made exactly 1 video of that quality in w .

Nonetheless, predicates of belief, unlike those of desire, lack the component in the meaning that would support the scalar presupposition of *even*. In other words, there is no context where the ordering obtains according to which John's belief of him making more than one video of that quality is more likely than his belief of him making exactly one video. Therefore, weak *even* is infelicitous under predicates of belief.

4.2. Covert *even* and weak REs

The parallelism between weak REs and weak *even* we have seen above suggests that the distribution of weak REs may be regulated by postulating an additional operator that carries out the same function as *even* does¹³. In the following, I use *EVEN* to represent this additional covert operator. With this assumption, examples where weak REs are not available may be accounted for in the same way as those where weak *even* is infelicitous. As far as I could see, there are two possibilities

¹³The assumption that AT-LEAST co-occurs with a covert *even* is empirically supported by the Slovenian concessive scalar particle *magari*, which is taken by Crnič (2011) to be the overt realization of (the combination of) AT-LEAST and *even*. As reported by Crnič (2011), *magari* may be glossed either as 'at least' or as 'even', and it is acceptable only if the inferences triggered by these two operators are consistent and satisfied in the context.

In the case of *believe* such as (90), EVEN triggers the scalar presupposition that some alternative in C” is more likely than the prejacent, namely *John’s mom believes that $SCORE(John) \geq SCORE(Bill)$ and it seems to her that $SCORE(John) = SCORE(Bill)$* .

(90) [EVEN-C” [EXH-C’ [*J’s mom* [*believe* [AT-LEAST-C [*John and Bill_F scored equally high*]]]]]

(91) $\llbracket (90) \rrbracket^w$ is defined only if:

$$\exists q \in C'' [\Box_{\text{dox}}(\mathbf{B}) \wedge \Diamond_{\text{dox}}(!\mathbf{B}) <_c q]; \quad (\text{Scalar presupposition})$$

if defined, $\llbracket (85a) \rrbracket^w = 1$ iff John’s mom believes that $SCORE(John) \geq SCORE(Bill)$ and it seems to her that $SCORE(John) = SCORE(Bill)$ in w.

believe/think, however, unlike predicates of desire, lacks the preference component in their meaning. Consequently, the scalar presupposition in (91) by no means can be satisfied, as we have seen in (87) for weak *even*. Hence, weak REs are not licensed under predicates of belief.

Along with the same lines, the license of weak REs in a modal statement like (5), which is uttered in the context like (6), may be captured as follows. (5) is parsed additionally with EVEN, as shown in (92). The prejacent of EVEN is presented with the shorthand $\Box(\mathbf{B}) \wedge \Diamond(!\mathbf{B})$ and is paraphrased as *it is necessary that $HEIGHT(\text{the-new-player}) \geq HEIGHT(\mathbf{B})$ and it is ok that $HEIGHT(\text{the-new-player}) = HEIGHT(\mathbf{B})$* . Along with these lines, EVEN triggers the scalar presupposition that some alternative in C” is more likely than the prejacent.

(92) [EVEN-C” [EXH-C’ [ModalP *have.to* [_{vP} AT-LEAST-C [*the-new-player and Bill_F*]] [AP *equally tall*]]]]

(93) $\llbracket (92) \rrbracket^w$ is defined only if:

$$\exists q \in C'' [\Box(\mathbf{B}) \wedge \Diamond(!\mathbf{B}) <_c q]; \quad (\text{Scalar presupposition})$$

if defined, $\llbracket (92) \rrbracket^w = 1$ iff

in contexts where the scalar presupposition of *even/EVEN* cannot be satisfied, embedded REs no longer carry an *at-least-equal-to* interpretation (i.e., they, as expected, only express strict mutual equivalence). I thank this reviewer for pointing this out. It seems that the prediction is borne out, at first glance, though further investigation is required. As the example below shows, assuming that typically it is preferable to pay less tax, the embedded RE can only express the mutual equivalence between the amounts of tax paid by Zhangsan and Lisi respectively.

- i. Zhāngsān₁ xīwàng qùnián tā jiǎo-dé shuì hé Lìsì yíyàng duō (jiùhǎo), zhèyàng tā₁
‘Zhangsan wish last.year he pay-REL tax AND Lisi equally many just this.way he
jīnnián jiù yǒu qián chūguó lè.
this.year then have money going.abroad PERF
‘Zhangsan wishes tht he paid the same amount of tax as Lisi did last year; this way he would
have money for going abroad.’

it is necessary that $\text{HEIGHT}(\text{the-new-player}) \geq \text{HEIGHT}(\text{B})$ and
it is OK that $\text{HEIGHT}(\text{the-new-player}) = \text{HEIGHT}(\text{B})$

In a context that supports goal-oriented modality (e.g., (6)), the alternatives in C” may be ordered based on the easiness these alternatives bear with respect to achieving the designated goal (i.e., p is more likely than q iff it is easier to reach the goal with p being true than with q being true). In such a context, the scalar presupposition may be satisfied if in the context it is easier to reach the goal (e.g., to be considered by the team/coach) with a height greater than that of Bill’s than with a height equivalent to that. Consequently, weak REs are licensed.

In summary, given the parallelism between weak *even* and weak REs in distribution, the license of weak REs, along with Crnič’s (2011) analysis, may be regulated by postulating a covert *even*; weak REs are acceptable in an embedding environment only if the scalar presupposition triggered by the covert *even* is satisfied in the context. Further investigation would be needed to see what predictions this analysis brings about and whether these predictions are borne out; this however would have to be left for future research.

5. Summary

In the discussion above I first present a novel observation that in Mandarin an RE, when embedded under a necessity modal or a predicate of desire, may have an $\geq$ -interpretation, which is weaker than how this degree construction may be interpreted when unembedded. I present an analysis of such examples and show that two covert operators may be at play: AT-LEAST serves to weaken the meaning of the embedded RE, and EXH, the exhaustification operator, gives rise to the inference of sufficiency. Along with Alonso-Ovalle and Hirsch, 2022, this analysis draws the connection between the examples investigated here and sufficiency modal constructions. Furthermore, along with Crnič’s (2011) analysis of weak *even*, I show that postulating a covert *even* may be a potential way to regulate the occurrence of this weakening operator and hence the distribution of weak REs.

One note before closing: there is one puzzle potentially related to the analysis proposed above. As a reviewer points out, while (94) carries the very same meaning as the Mandarin counterpart of *John and Bill scored equally high*, a desire report with (94) embedded (see (95)), in contrast to (10), does not seem to have the $\geq$ -interpretation.

- (94) Yuēàn hé Bǐěr kǎo-lè yímóyíàng-dè fēnshù.
John AND Bill score-PERF exactly.identical-MOD score
‘John and Bill got exactly identical score.’

- (95) Yūehàn₁-dè māmā xīwàng tā₁ hé Bǐěr kǎo yímóyíàng-dè fēnshù.
John-POSS mother wish he AND Bill score exactly.identical-MOD score

At this moment I do not have anything smart to say and would have to leave it for future investi-

gation, which perhaps need to be centered on the difference between *identical* and equivalence on degree comparison; after all, unlike REs, *identical* does not necessarily encode gradability.

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References

- Alonso-Ovalle, L., & Hirsch, A. (2022). Keep *only* strong. *Semantics and Pragmatics*, 15(6). <https://doi.org/https://doi.org/10.3765/sp.15.6>. This
- Bennett, J. (1982). *Even if*. *Linguistics and Philosophy*, 5, 403–418.
- Chierchia, G. (2004). Scalar implicatures, polarity phenomena, and the syntax/pragmatics interface. In A. Belletti (Ed.), *Structures and beyond: The cartography of syntactic structures* (pp. 39–103, Vol. 3). Oxford University Press.
- Chierchia, G. (2013). *Logic in grammar: Polarity, free choice, and intervention*. Oxford University Press.
- Chierchia, G., Fox, D., & Spector, B. (2012). Scalar implicature as a grammatical phenomenon. In C. Maieborn, K. von Steubner, & P. Portner (Eds.), *Semantics: An international handbook of natural language meaning* (pp. 2297–2331, Vol. 3). Mouton de Gruyter.
- Chomsky, N. (1993). A minimalist program for linguistic theory. In K. Hale & S. J. Keyser (Eds.), *The view from building 20: Essays in linguistics in honor of sylvain bromberger* (pp. 1–52). MIT.

- Condoravdi, C., & Francez, I. (2022). A minimally sufficient analysis of sufficiency modal constructions. *Proceedings of the 23rd Amsterdam Colloquium*, 77–83. <https://www.dropbox.com/s/qv9qzppggxz23er/Proceedings2022-pages-84-90.pdf?dl=0>
- Coppock, E., & Beaver, D. I. (2013). Principles of the Exclusive Muddle. *Journal of Semantics*, 31(3), 371–432. <https://doi.org/10.1093/jos/fft007>
- Crnič, L. (2011). *Getting even* [Doctoral dissertation, Massachusetts Institute of Technology]. <https://doi.org/http://doi.org/1721.1/68912>
- Crnič, L. (2014). Non-monotonicity in npi licensing. *Natural Language Semantics*, 22(2), 169–217. <https://doi.org/10.1007/s11050-014-9104-6>
- Crnič, L. (2019). Any: Logic, likelihood, and context (pt. 2) [e12353 10.1111/lnc3.12353]. *Language and Linguistics Compass*, 13(11), e12353. <https://doi.org/https://doi.org/10.1111/lnc3.12353>
- Erlewine, M. Y. (2014). Explaining leftward focus association with even but not only. In U. Etxeberria, A. Făalăuș, A. Irurtzun, & B. Leferman (Eds.), *Proceedings of sinn und bedeutung 18*.
- Erlewine, M. Y. (2018). Clausal comparison without degree abstraction in mandarin chinese. *Natural Language and Linguistic Theory*, 36, 445–482.
- Fintel, K. von. (1999). NPI licensing, Strawson Entailment, and Context Dependency. *Journal of Semantics*, 16(2), 97–148.
- Fintel, K. von, & Iatridou, S. (2007). Anatomy of a modal construction. *Linguistic Inquiry*, 38(3), 445–483.
- Fox, D. (1999). Reconstruction, binding theory, and the interpretation of chains. *Linguistic Inquiry*, 30(2), 157–196. <https://doi.org/10.1162/002438999554020>
- Fox, D. (2002). Antecedent-contained deletion and the copy theory of movement. *Linguistic Inquiry*, 33(1), 63–96. <https://doi.org/10.1162/002438902317382189>
- Fox, D. (2007). Free choice disjunction and the theory of scalar implicatures. In U. Sauerland & P. Stateva (Eds.), *Presupposition and implicature in compositional semantics*. Palgrave Macmillan.
- Fox, D., & Katzir, R. (2011). On the characterization of alternatives. *Natural Language Semantics*, 19(1), 87–107. <https://doi.org/10.1007/s11050-010-9065-3>
- Guerzoni, E. (2003). *Why even ask? on the pragmatics of questions and the semantics of answers* [Doctoral dissertation, Massachusetts Institute of Technology].

- Guerzoni, E. (2004). Even-NPIs in YES/NO questions. *Natural Language Semantics*, 12(4), 319–343. <https://doi.org/10.1007/s11050-004-8739-0>
- Heim, I. (1984). A note on negative polarity and downward entailingness. In C. Jones & P. Sells (Eds.), *Proceedings of the 14th north eastern linguistic society meeting* (pp. 98–107). GLSA.
- Horn, L. R. (1969). A presuppositional analysis of only and even. *Proceedings from the Annual Meeting of the Chicago Linguistic Society*, 5(1), 98–107.
- Horn, L. R. (1996). Exclusive company: Only and the dynamics of vertical inference. *Journaltitle of Semantics*, 13(1), 1–40. <https://doi.org/10.1093/jos/13.1.1>
- Hsieh, I.-T. C. (2015). Long-distance reflexives, blocking effects, and the structure of mandarin comparatives. *Syntax*, 18(1), 78–102. <https://doi.org/10.1111/synt.12026>
- Hsieh, I.-T. C. (2017). On arguments against comparative deletion in mandarin. *Tsing Hua Journaltitle of Chinese Linguistics*, 47, 255–287.
- Hsieh, I.-T. C. (2021). On reciprocal degree constructions: A view from mandarin. *Language and Linguistics*, 22(2), 272–301.
- Jackendoff, R. . (1972). *Semantic interpretation in generative grammar*. MIT Press. <https://books.google.com.tw/books?id=JwvuSgAACAAJ>
- Karttunen, L., & Peters, S. (1979). Conventional implicature. In C.-K. Oh & D. A. Dinneen (Eds.), *Syntax and semantics* (pp. 1–56, Vol. 11). Academic Press.
- Katzir, R. (2007). Structurally-defined alternatives. *Linguistics and Philosophy*, 30(6), 669–690. <https://doi.org/10.1007/s10988-008-9029-y>
- Kay, P. (1990). Even. *Linguistics and Philosophy*, 13, 59–111.
- Krasikova, S. (2010). Sufficiency inference in anankastic conditionals. *Semantics and Linguistic Theory*, 20, 91–108.
- Lahiri, U. (1998). Focus and negative polarity in hindi. *Natural Language Semantics*, 6(1), 57–123. <https://doi.org/10.1023/A:1008211808250>
- Nissenbaum, J., & Schwarz, B. (2011). Parasitic degree phrases. *Natural Language Semantics*, 19(1), 1–38.
- Paris, M.-C. (2008). On parts of speech in chinese: *gen*. *The Linguistic Review*, 25, 347–366.
- Schwarz, B. (2005). Scalar additive particles in negative contexts. *Natural Language Semantics*, 13, 125–168.

Schwarz, B. (2007). Reciprocal equatives. In E. Puig-Waldmüller (Ed.), *Proceedings of sinn und bedeutung 11*. (pp. 568–582). Universitat Pompeu Fabra. <http://www.upf.edu/dtf/sub11>

Zhang, N. N. (2010). *Coordination in syntax*. Cambridge University Press.

Address for correspondence

I-Ta Chris Hsieh

Graduate Institute of Linguistics

National Tsing Hua University, Taiwan

101, Kuang-Fu Road, Sec. 2, Hsinchu 300044, Taiwan

ita.hsieh@mx.nthu.edu.tw

ORCID: <https://orcid.org/0000-0001-9493-5728>